

PREDICTING VEGETATION RECOVERY POST-FIRE

Kate McCarthy

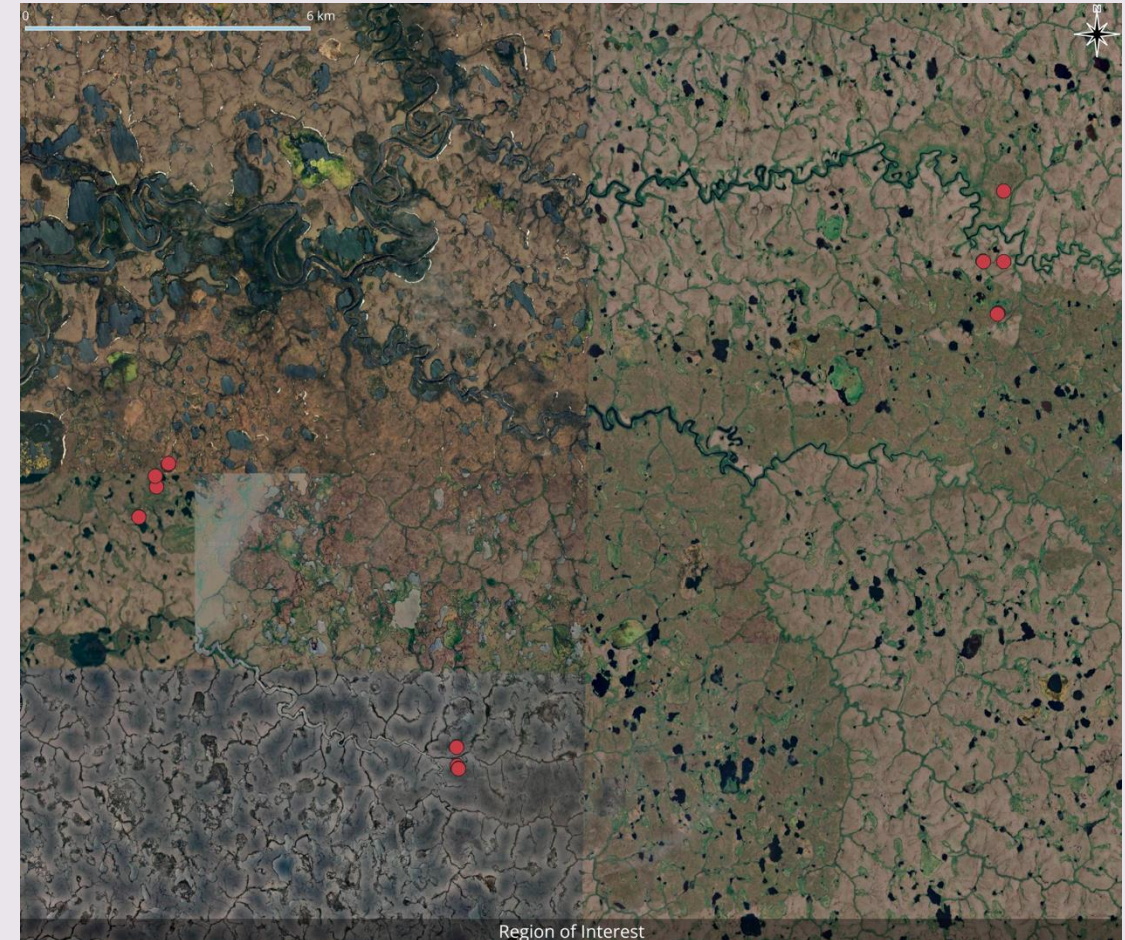
Dr. Jessica O'Connell

Reproducible Remote Sensing Models

11 March 2026

Introduction

- Fires have been increasing in number and severity due to anthropogenic climate change (Melvin et al., 2017).
- Previous models built on earlier data need to be updated to be more accurate for vegetation recovery in a changing climate.
- Study Site: Yukon-Kuskokwim Delta (YKD in Alaska)
 - *One of the most biologically productive areas in the Arctic Tundra (Frost et al., 2020)*
 - *17 known fires since 1950, 7 of them in the 21st century (Frost et al., 2020)*
- Problem: Vegetation recovery is hard to track in remote places with in-situ data; using a remote sensing model can help predict recovery following fires over a larger area.



Context & Objectives

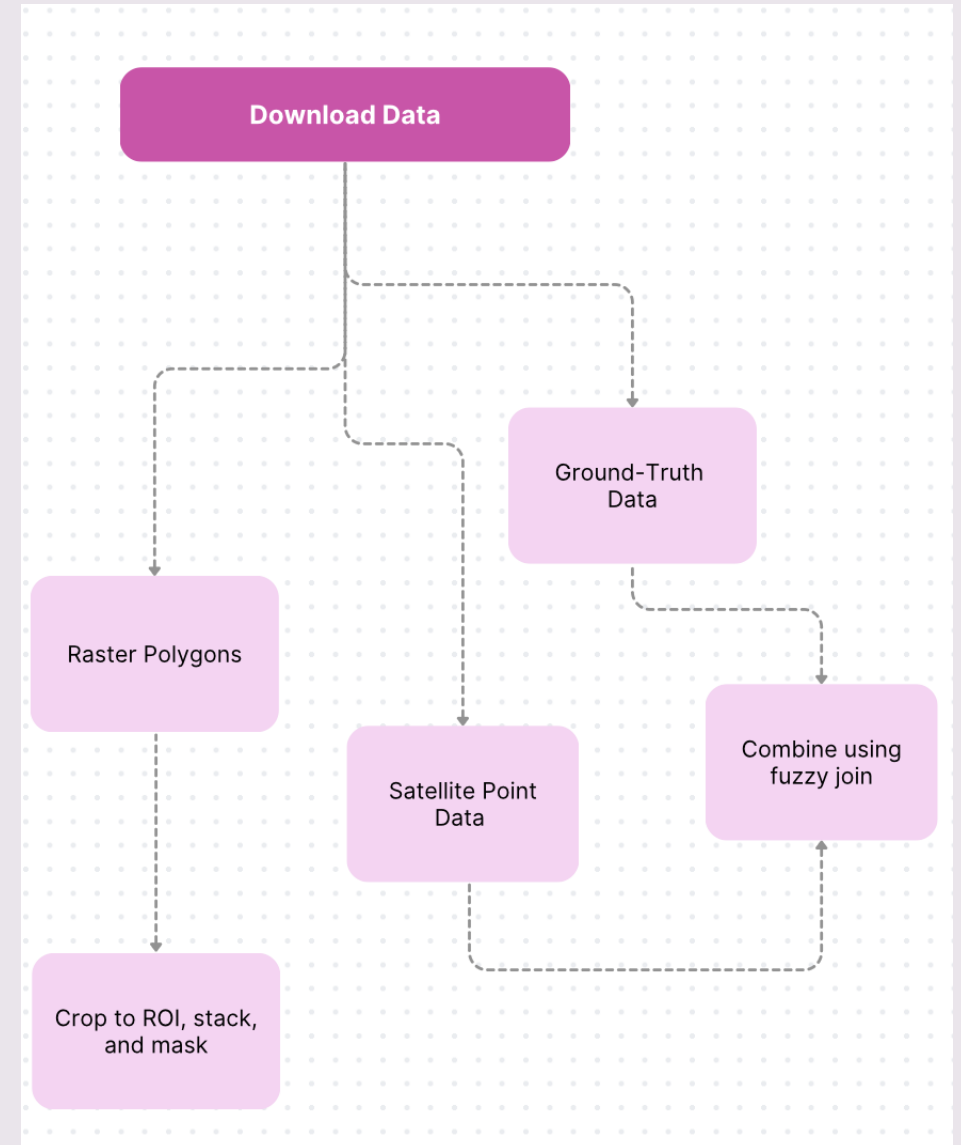
- Spatiotemporal scope:
 - *Latitude: 61.2361 to 61.3455*
 - *Longitude: -163.3481 to -163.0079*
 - *8 years, spanning from pre-fire (2014) to 7-years post-fire (2022)*
- Environmental Context:
 - *Recent fires in 2015 (Hung et al, 2022; Frost et al., 2020)*
 - *Relatively flat area of wetlands, peat plateaus, streams, and lakes (Hung et al., 2022)*
- Objective: Track vegetation recovery following a fire in the YKD in Alaska.
 - *Combine ground-truth and raster data for 2019*
 - Use this to build model to predict above-ground biomass (AGBio)
 - *Compare across time (2014, 2016, 2019, 2022) to track vegetation regrowth*



Medici, 2016 (Courtesy of Calista Education and Culture, Inc.)

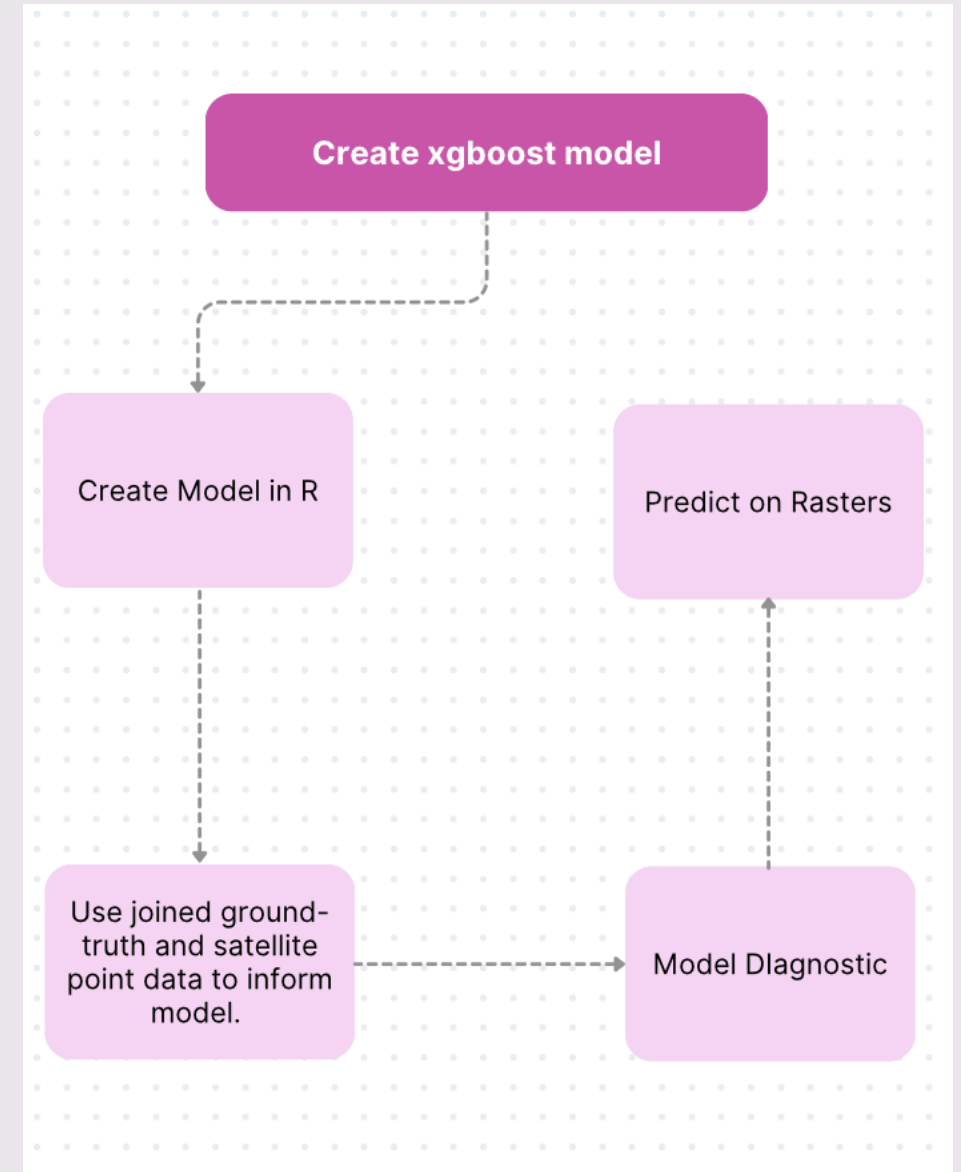
Methods: Data Gathering & Processing

- Used published data from the Polaris Project 2019 for ground-truth data of above-ground vegetation biomass in the Yukon-Kuskokwim Delta, AK (Hung et al., 2022)
- Used APPeears API to download HLS raster (polygon) data from Landsat 8/9 and Sentinel-2 satellites from 2014, 2016, 2019, and 2022, and HLS point data from 2019 to train the model.
 - *Blue, Green, Red, NIR, SWIR1, SWIR2 bands*
 - *Polygon: cropped rasters to region of interest*
- Prior to modelling: cleaned polygon, point, ground-truth data, combined point and ground-truth datasets using a fuzzy join, and stacked rasters.



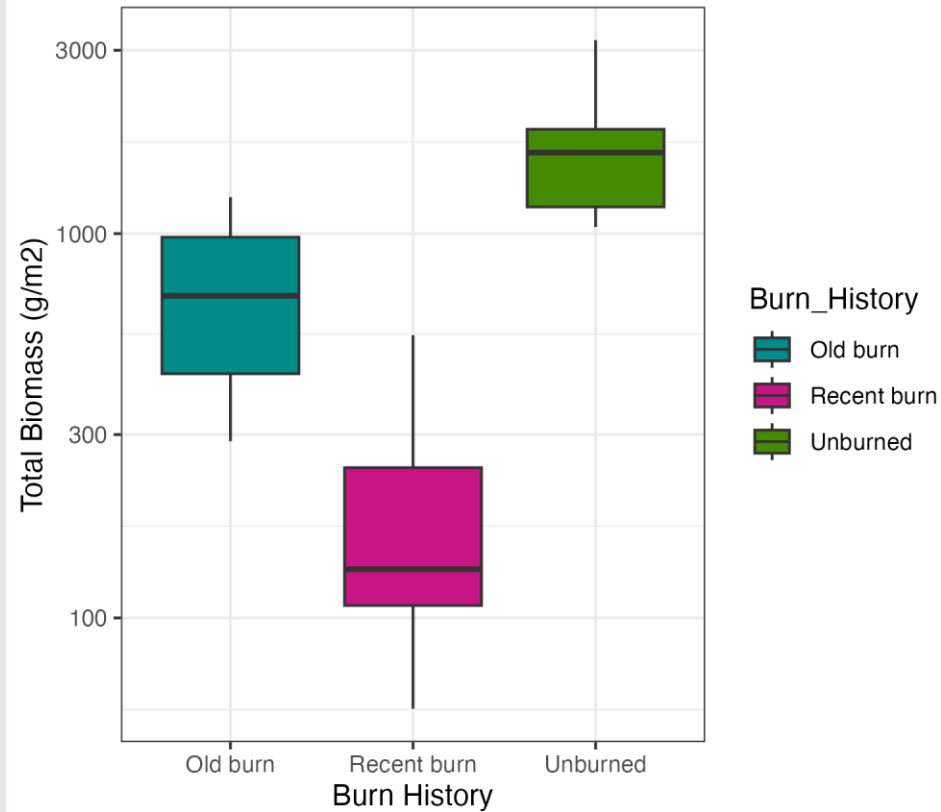
Methods: Model Building

- Built an XG Boost regression model to predict AGBio from the reflectance and ground-truth joined dataset
 - 500 iterations
 - Created 3 models and averaged
- Indices:
 - $NDVI = (NIR - Red) / (NIR + Red)$
 - Normalized Difference Vegetation Index
 - Has been used to examine vegetation recovery after fire (Lacouture et al., 2020)
 - $NMDI = (NIR - SWIR) / (NIR + SWIR)$
 - Normalized Multiband Drought Index
 - Used to detect vegetation water content (USGS, n.d.)
 - $VARI = (Green - Red) / (Green + Red - Blue)$
 - Visible Atmospherically Resistant Index
 - Used for to measure chlorophyll level in vegetation (Solvi, 2023)



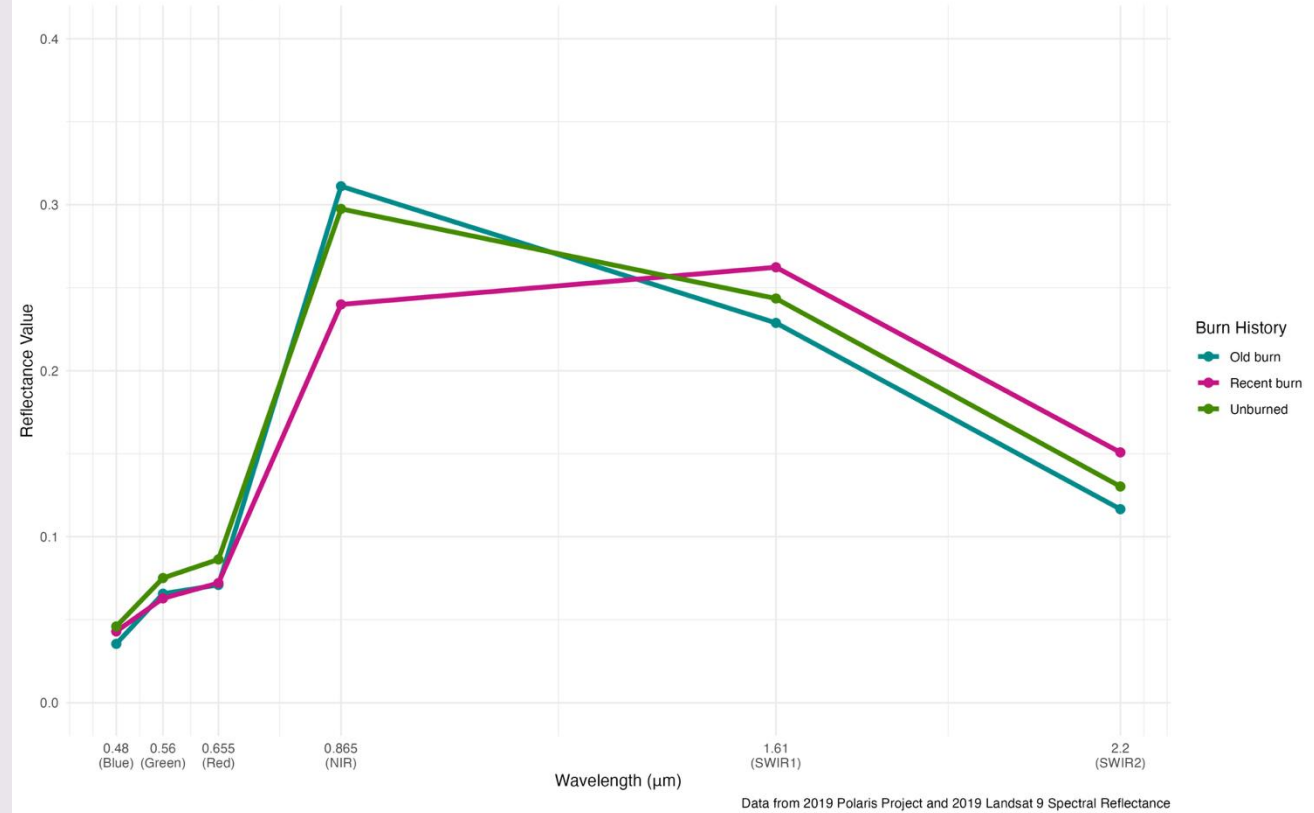
Results: Data-Informing the Model

Biomass Comparison by Burn History
in the Yukon-Kuskokwim Delta in Alaska



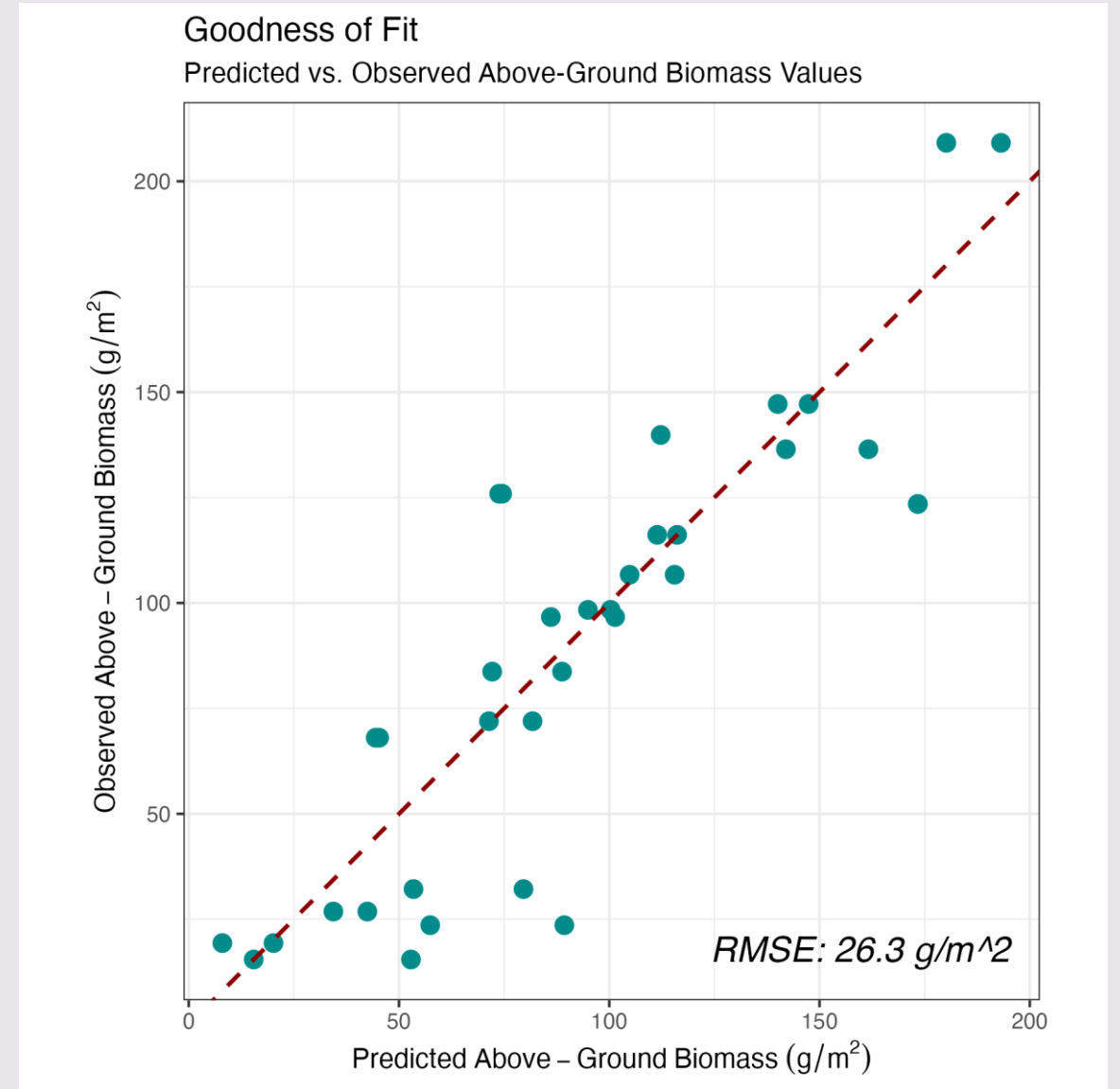
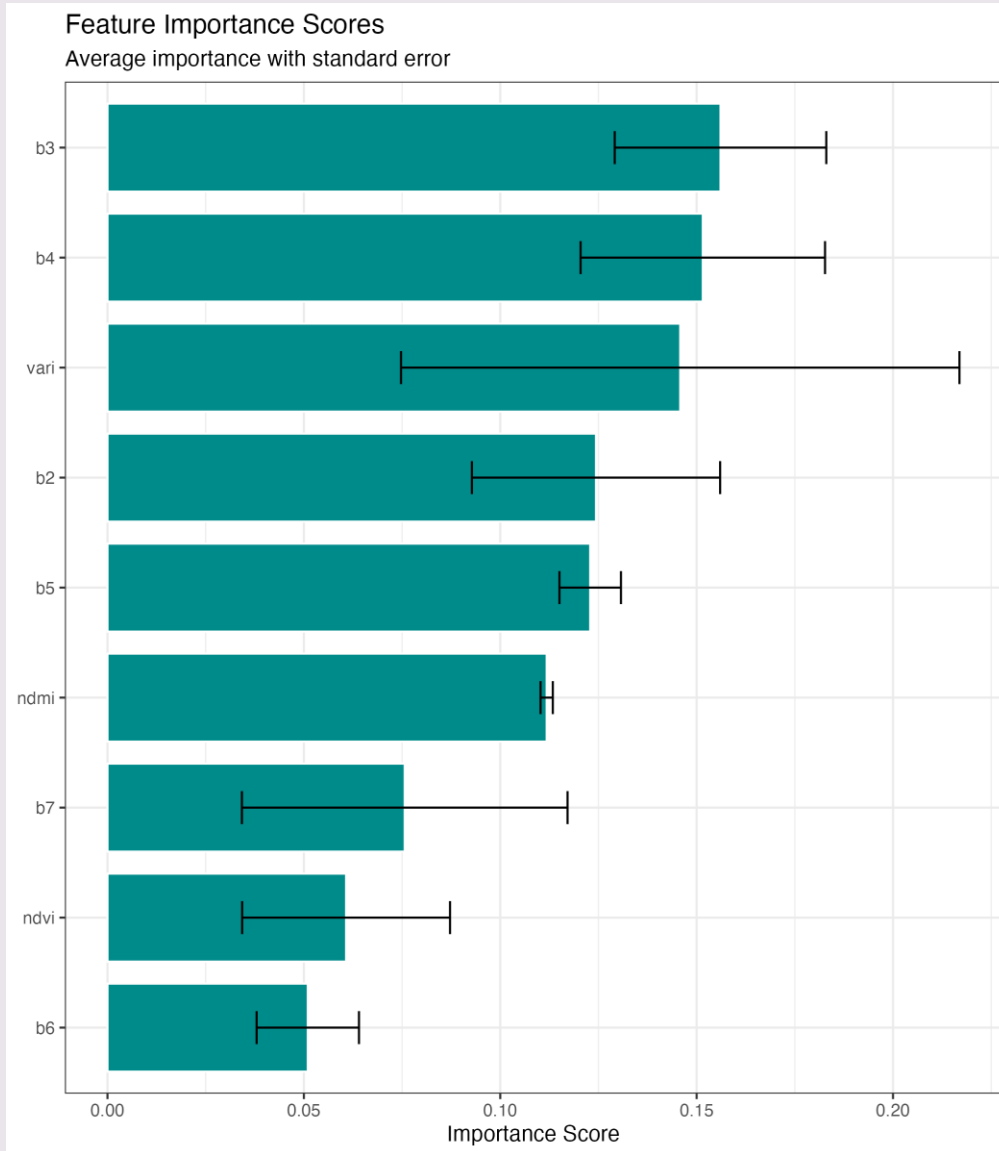
Data was adapted from Polaris Project: 2019 by Hung et al., 2022.

Spectral Reflectance by Burn History
in the Yukon-Kuskokwim Delta in Alaska



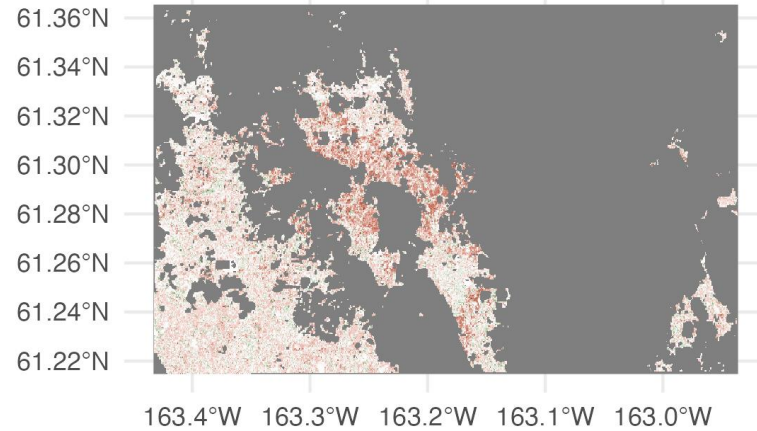
Data from 2019 Polaris Project and 2019 Landsat 9 Spectral Reflectance

Results: Model Diagnostics & Goodness of Fit

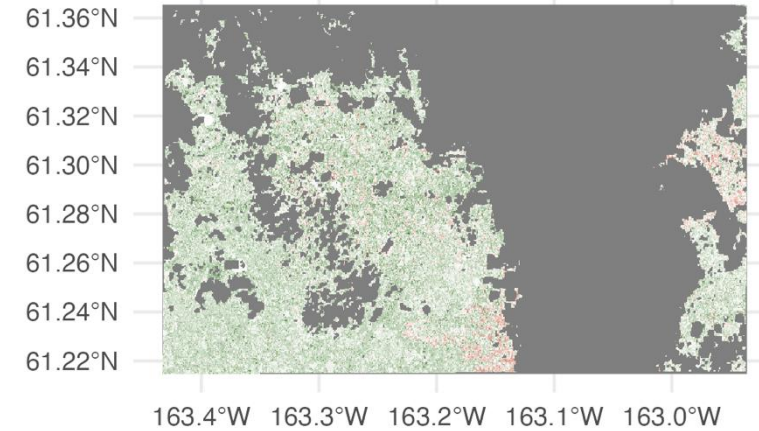


Results: Model Predictions

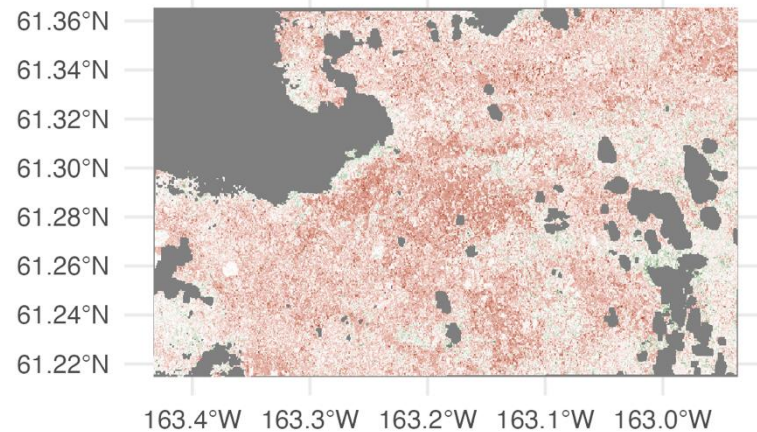
Change in Biomass: 2014 to 2016



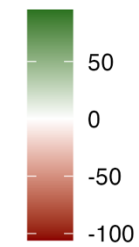
Change in Biomass: 2016 to 2019



Change in Biomass: 2019 to 2022

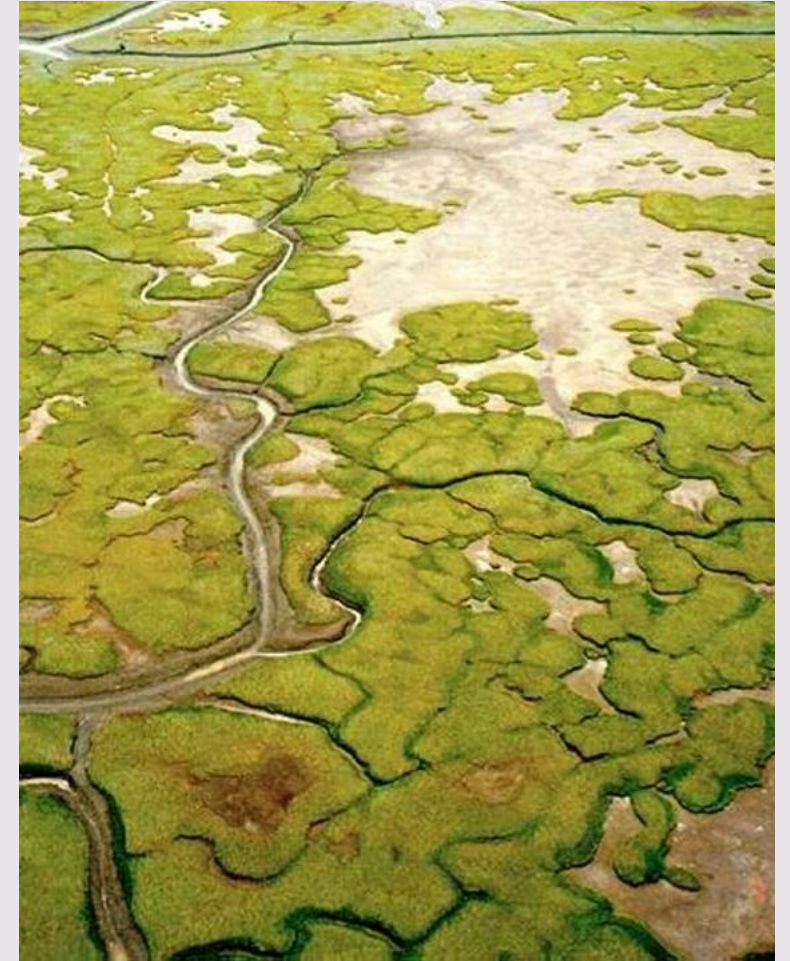


Change in Biomass (g/m²)



Discussion/Conclusion:

- Successfully implemented workflow from beginning to end in R
- Spectral indices captured vegetation health and differences across time.
- Root mean squared error of 26.3 on a scale of 200.
- Vegetation health in 2022 appears to be lower than earlier years, which contradicts estimated vegetation recovery.
 - *Fires ransacked Alaska in 2022, including in the YKD (Rosen, 2022).*
- Found 2022 ground-truth data from the same group (Polaris Project: 2022)
 - *Could be used to better train the model.*
 - *Larger sample size to create split groups and validation set.*



Alaska Ecoscience, n.d.

Sources:

Ecological monitoring Yukon-Kuskokwim Delta. (n.d.). Alaska Ecoscience. <https://alaskaecoscience.com/projectsecological-monitoring-yk-delta/>

Frost, G. V., Loehman, R. A., Saperstein, L. B., Macander, M. J., Nelson, P. R., Paradis, D. P., & Natali, S. M. (2020). *Multi-decadal patterns of vegetation succession after tundra fire on the Yukon-Kuskokwim Delta, Alaska*. *Environmental Research Letters*, 15(2), 25003.

<https://doi.org/10.1088/1748-9326/ab5f49>

Jacqueline Hung, Susan Natali, Natalie Baillargeon, Seeta Sistla, Grace Pold, & Rhys MacArthur. (2022). *Polaris Project 2019: Vegetation biomass, point intercept, and thaw depth, Yukon-Kuskokwim Delta, Alaska*. Arctic Data Center. [doi:10.18739/A2JS9H89M](https://doi.org/10.18739/A2JS9H89M).

Lacouture, D. L., Broadbent, E. N., & Crandall, R. M. (2020). *Detecting Vegetation Recovery after Fire in A Fire-Frequented Habitat Using Normalized Difference Vegetation Index (NDVI)*. *Forests*, 11(7), 749. <https://doi.org/10.3390/f11070749>

Medici, L. (2016, June 9). *Slings, arrows, and deceit in the western reaches of Alaska*. National Endowment for the Humanities. <https://www.neh.gov/divisions/research/featured-project/slides-arrows-and-deceit-in-the-western-reaches-alaska-traditional-tales-the-yup%E2%80%99ik-people>

Melvin, A. M., Murray, J., Boehlert, B., Martinich, J. A., Rennels, L., & Rupp, T. S. (2017). *Estimating wildfire response costs in Alaska's changing climate*. *Climatic Change*, 141(4), 783 – 795. <https://doi.org/10.1007/s10584-017-1923-2>

Normalized difference moisture index. (n.d.) USGS. <https://www.usgs.gov/landsat-missions/normalized-difference-moisture-index>

Rosen, Y. (2022, August 15). *Tundra Burns helped make the 2022 Alaska fire season one of the biggest since 1950*. Alaska Beacon. <https://alaskabeacon.com/briefs/tundra-burns-helped-make-the-2022-alaska-fire-season-one-of-the-biggest-since-1950/>

Solvi. (2023, March 1). *Which vegetation index should I use?*. Solvi Blog. <https://solvi.ag/blog/which-vegetation-index-should-i-use/>